

PROBABILITY AND STATISTICS

Probability:

Distribution $\longrightarrow$ Samples

Given distribution find probabilities of data/events.

Ex: $X \sim \text{Hypergeometric}(r = 70, b = 30, n = 10)$

The probability of a random sample of size n having 5 red balls?

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Statistics:

Sample $\longrightarrow$ Distribution

Given data find parameters/properties of distribution.

Ex: We observed $X_1 = 0, X_2 = 1, \dots, X_{10} = 0$

What is the distribution parameter p , i.e. probability of heads?

POINT ESTIMATION

Find single “good estimate” of a quantity of interest of a distribution/population using statistics.

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Formally, $X_1, \dots, X_n$ iid data points from some distribution.

An *estimate* of parameter θ of the distribution: $\hat{\theta}_n = r(X_1, \dots, X_n)$

for some appropriate function r .

Note: θ : fixed, unknown quantity. $\hat{\theta}_n$: random variable.

POINT ESTIMATION

Ex: Estimate $\theta = \mu = \sum_x x f(x)$ of an unknown distribution.

Say true (unknown) value $\theta = 3.5$.

Sample 4 data points X_1, X_2, X_3, X_4 , say 3, 6, 5, -2 .

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Can try to estimate θ with **any** function of $X_1, \cdots, X_4$:

$\hat{\theta}_n:$

$\frac{X_1 + \cdots + X_n}{n}$	$\frac{\min(X_1, \cdots, X_n) + \max(X_1, \cdots, X_n)}{2}$	$X_1 \cdot X_n$
3	2	-6

PLUG-IN ESTIMATOR

Property of distribution: θ

Min: $\min_{\nabla_x} \{x : f(x) > 0\} \longrightarrow$

Plug-in sample: $\hat{\theta}_n$

Sample min: $\min(X_1, \dots, X_n)$

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Min: $\min_{\forall x} \{x : f(x) > 0\} \longrightarrow$

Mean: $\mu = \sum_x x f(x) \longrightarrow$

Correlation: $\longrightarrow$

$$\rho(X, Y) = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sqrt{(E[(X - \mu_X)^2])} \sqrt{(E[(Y - \mu_Y)^2])}}$$

Plug-in sample: $\hat{\theta}_n$

Sample min: $\min(X_1, \dots, X_n)$

Sample mean: $\frac{1}{n} \sum_{i=1}^n X_i$

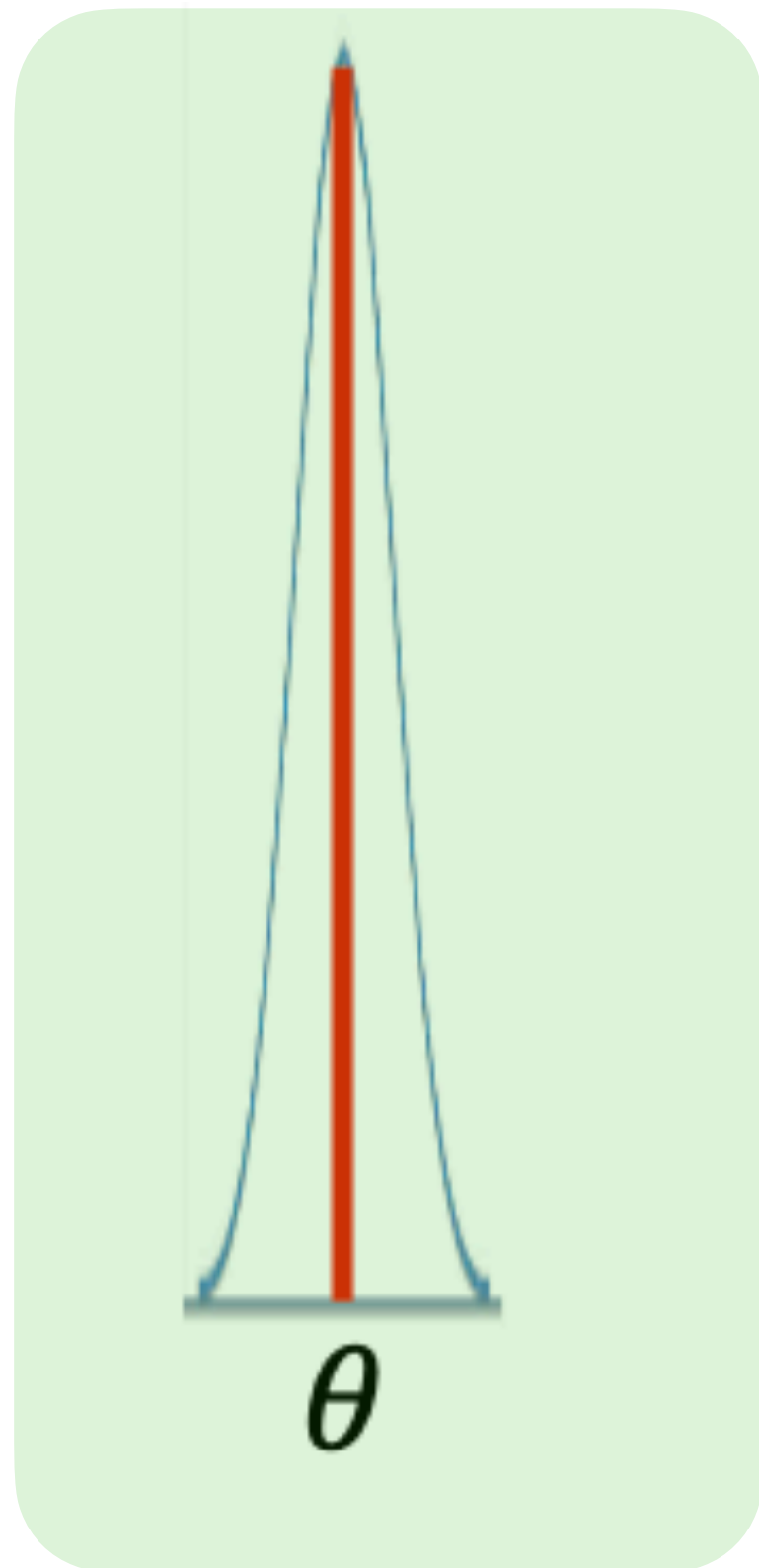
Sample Correlation:

$$\frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

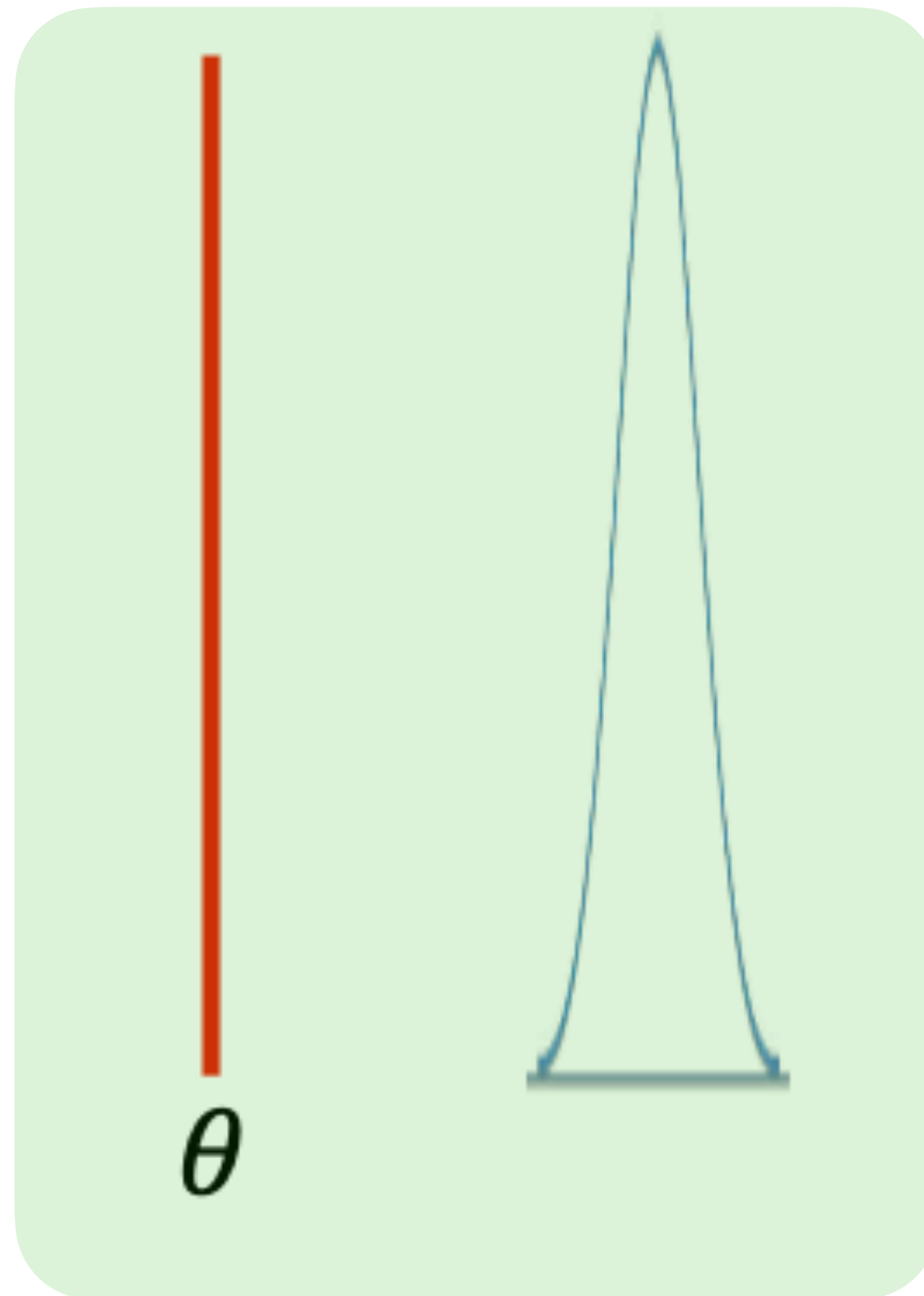
HOW GOOD IS AN ESTIMATOR?

Several possible estimators. Quality of an estimator?

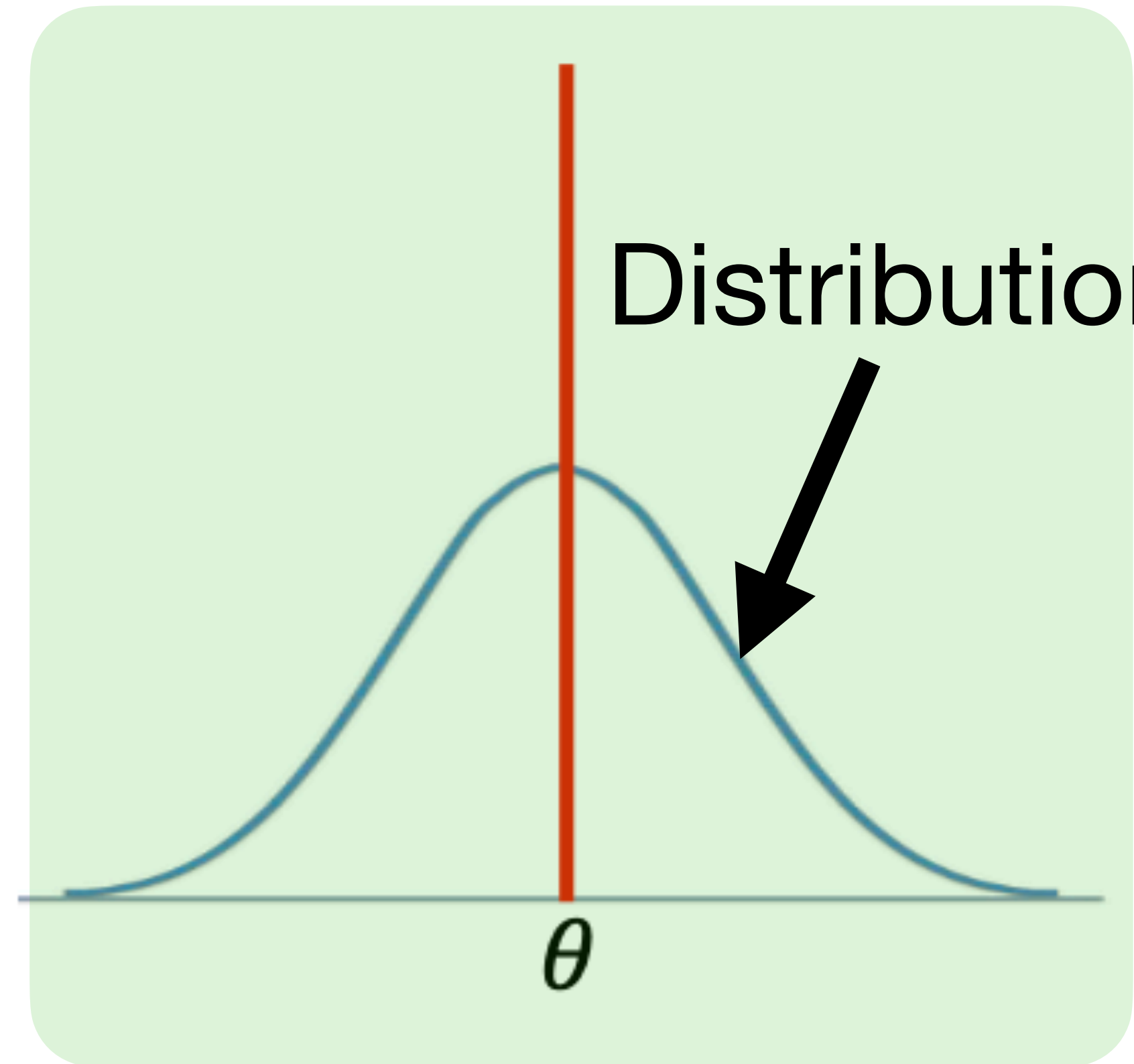
Good



Low bias
Low variance



Bias

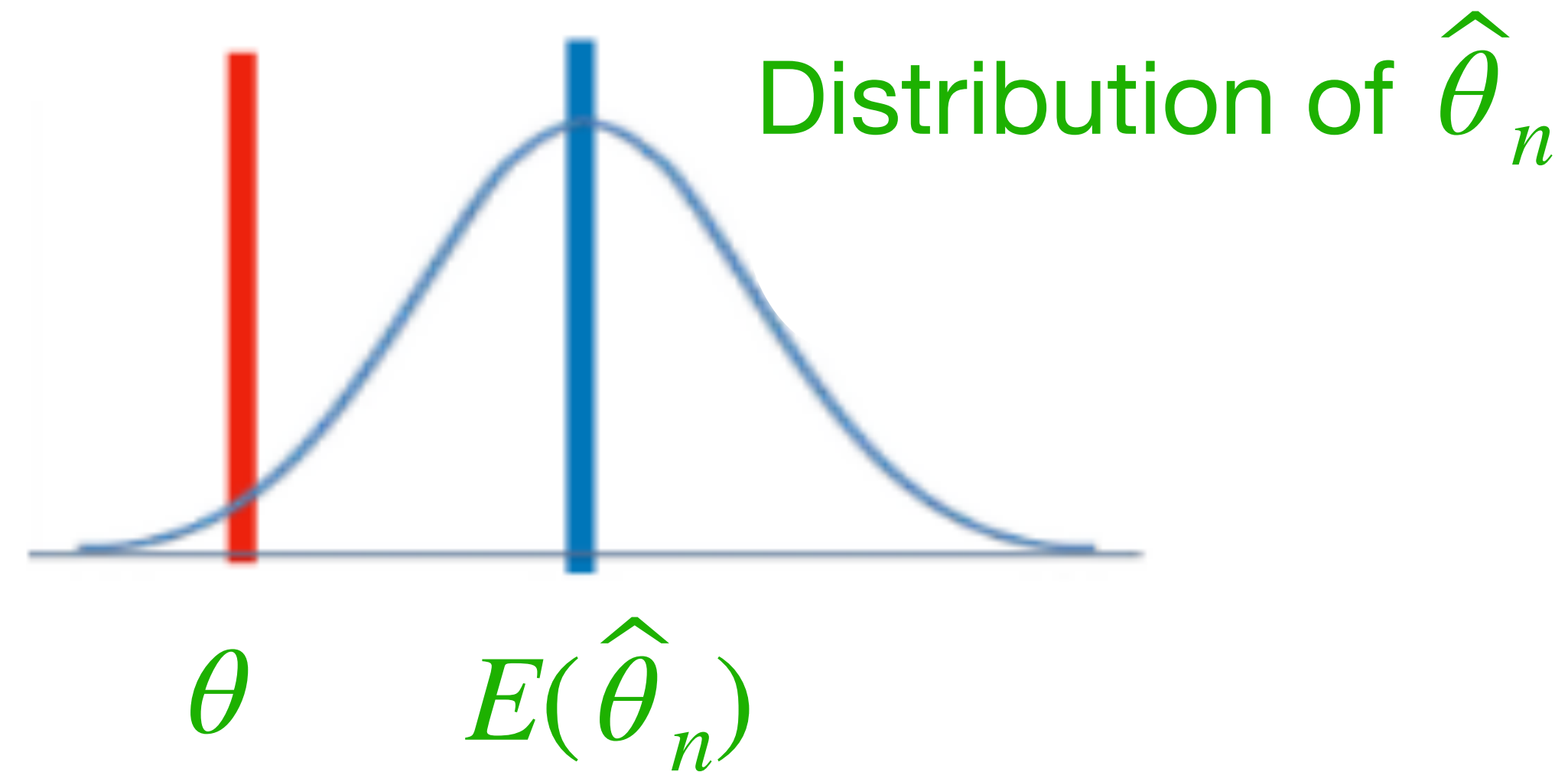


Variance

BIAS, VARIANCE

Bias: Expected overestimate of θ :

$$\text{Bias}(\hat{\theta}_n) = E(\hat{\theta}_n - \theta) = E(\hat{\theta}_n) - \theta$$

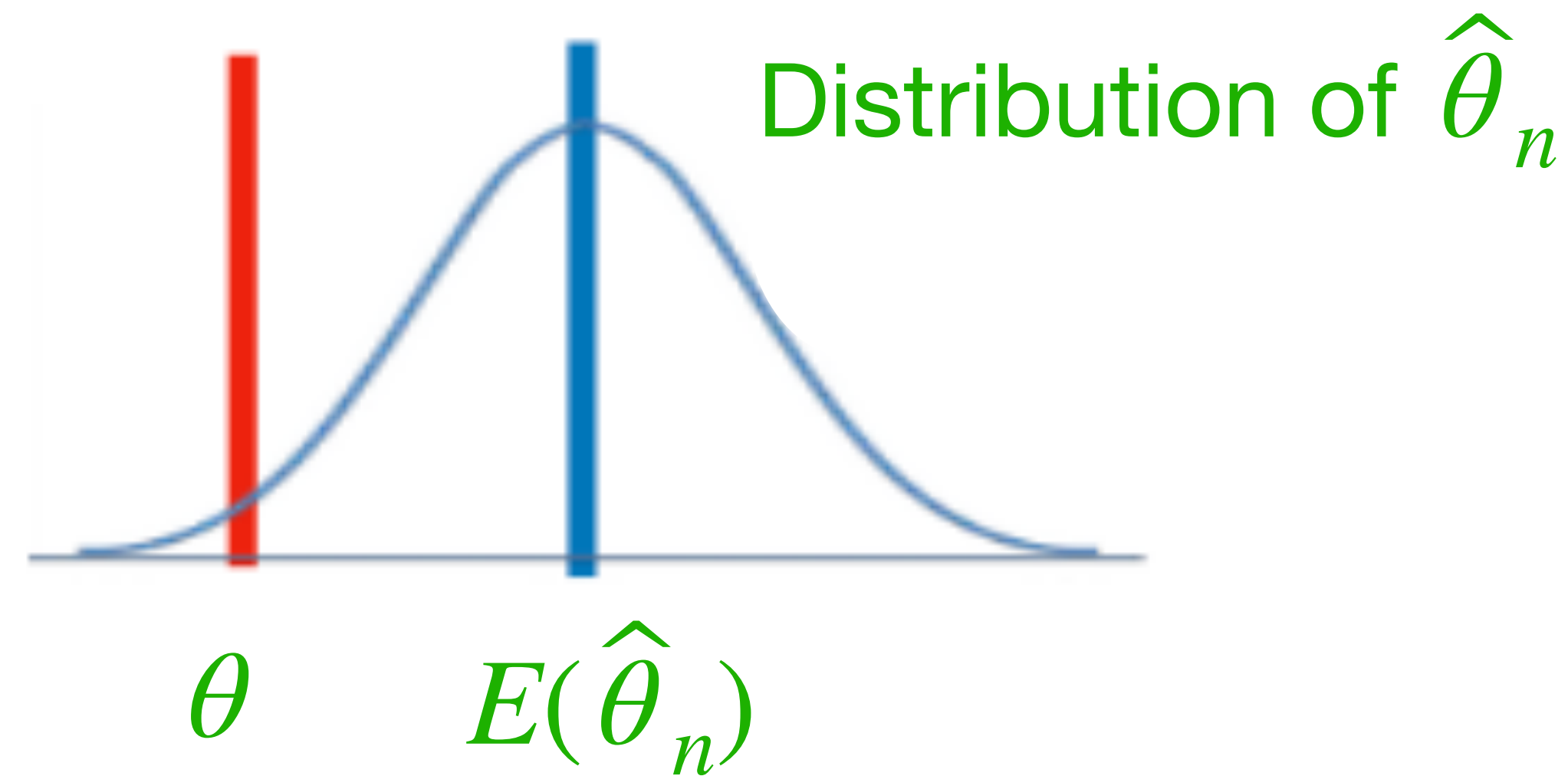


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Variance: $\text{Var}(\hat{\theta}_n) = E[(\hat{\theta}_n - E(\hat{\theta}_n))^2]$

Bias-Variance tradeoff is typical.

BIAS, VARIANCE

Ex: Observe n coin flips $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

True value of p unknown. Want to estimate it.

Possible estimator: Sample mean $\bar{X}_n = \frac{1}{n} \sum X_i$

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Recall: $E(\bar{X}_n) = \frac{1}{n} \sum E(X_i) = p,$

Sample mean is an unbiased estimator.

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum \text{Var}(X_i) = \frac{p(1-p)}{n}$$

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Quality of an estimator $\hat{\theta}_n$:

$$\text{mean squared error (mse)} = E[(\hat{\theta}_n - \theta)^2]$$

Theorem: $mse = bias^2(\hat{\theta}_n) + Var(\hat{\theta}_n)$

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Proof:

$$\begin{aligned} E[(\hat{\theta}_n - \theta)^2] &= E[(\hat{\theta}_n - \mu_{\hat{\theta}_n} + \mu_{\hat{\theta}_n} - \theta)^2] \\ &= E[(\hat{\theta}_n - \mu_{\hat{\theta}_n})^2] + E[(\mu_{\hat{\theta}_n} - \theta)^2] + 2E[\hat{\theta}_n - \mu_{\hat{\theta}_n}]E[\mu_{\hat{\theta}_n} - \theta] \end{aligned}$$

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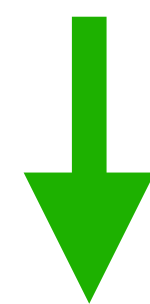
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$$Var(\hat{\theta}_n)$$



$$(\mu_{\hat{\theta}_n} - \theta)^2 = Bias^2(\hat{\theta}_n)$$

BIAS-VARIANCE TRADEOFF

Ex: Observe n coin flips $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

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Found: $E(\bar{X}_n) = p, \quad \text{Var}(\bar{X}_n) = \frac{p(1-p)}{n} \quad \Rightarrow \quad \text{mse} = \frac{p(1-p)}{n}$

$\bar{X}_n$ is unbiased estimator, call it $\hat{\theta}_U$.

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Ex: Observe n coin flips $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

True value of p unknown. Want to estimate it.

Alternative estimator, call it $\hat{\theta}_B$: $\frac{1 + \sum_{i=1}^n X_i}{n + 2}$

e.g., 7 successes out of 10 trials, estimate of p with:

$$\hat{\theta}_B : 8/12 \approx 0.67, \quad \hat{\theta}_U : \frac{7}{10} = 0.7$$

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Bias of $\hat{\theta}_B$: $\frac{1 + np}{n + 2} - p$, **biased**.

Variance of $\hat{\theta}_B$: $\frac{1}{(n + 2)^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{np(1 - p)}{(n + 2)^2}$

BIAS-VARIANCE TRADEOFF

Ex: Observe n coin flips $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

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Which one is a better estimator for $n = 10$?

$$\text{mse of } \hat{\theta}_U: \frac{p(1-p)}{10}, \quad \text{mse of } \hat{\theta}_B: \frac{-6p^2 + 6p + 1}{144}$$

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